

Ultrasonic Testing and Velocity Measurement, Material Prediction

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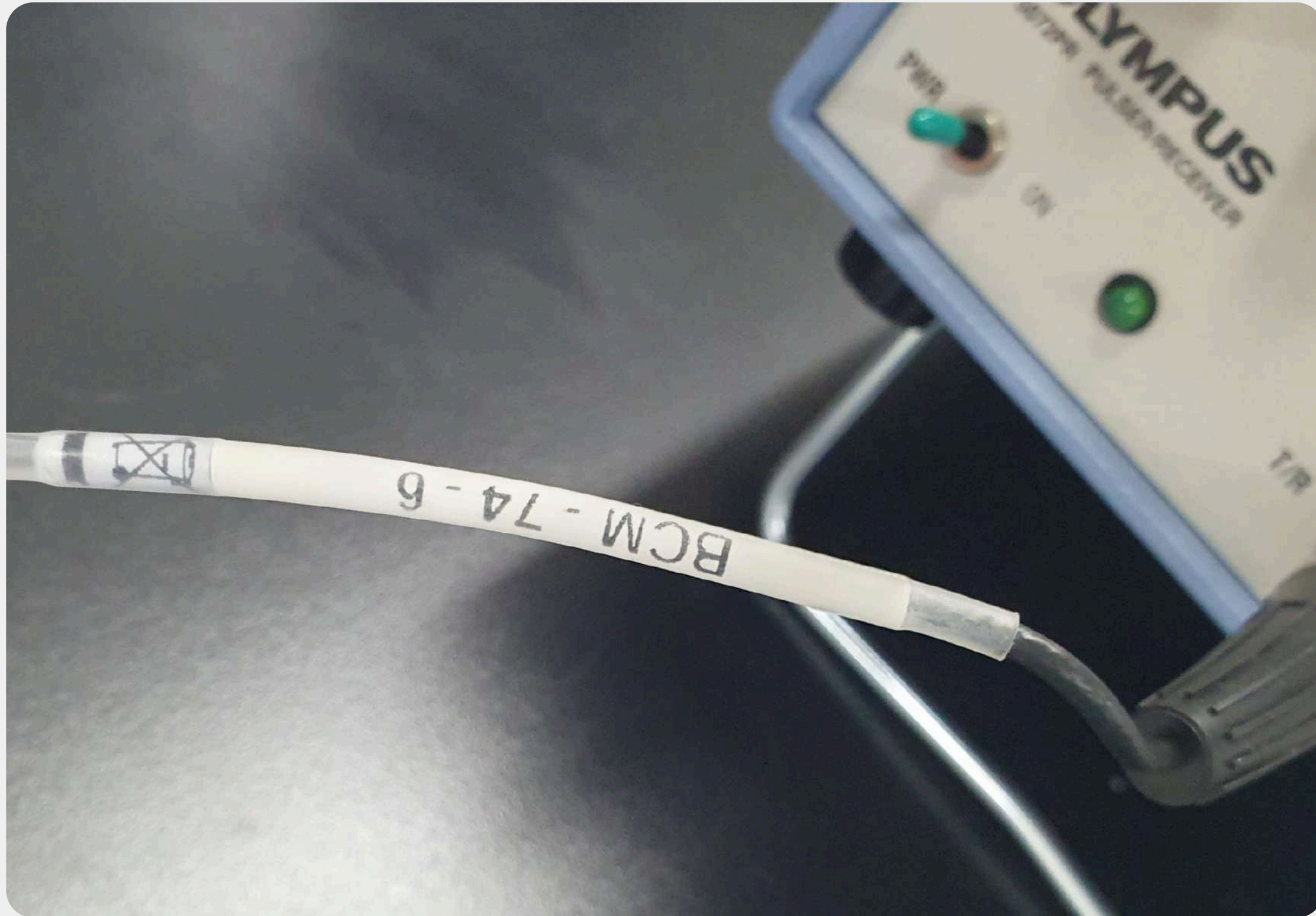
- 표면이 매끄럽다.
- 단단하지만 금속의 재료는 아닌 것 같다.
- 첫번째 반사 이후 두번째 반사신호를 받기 어렵다.
- 즉, Damping이 높은 시편이다.

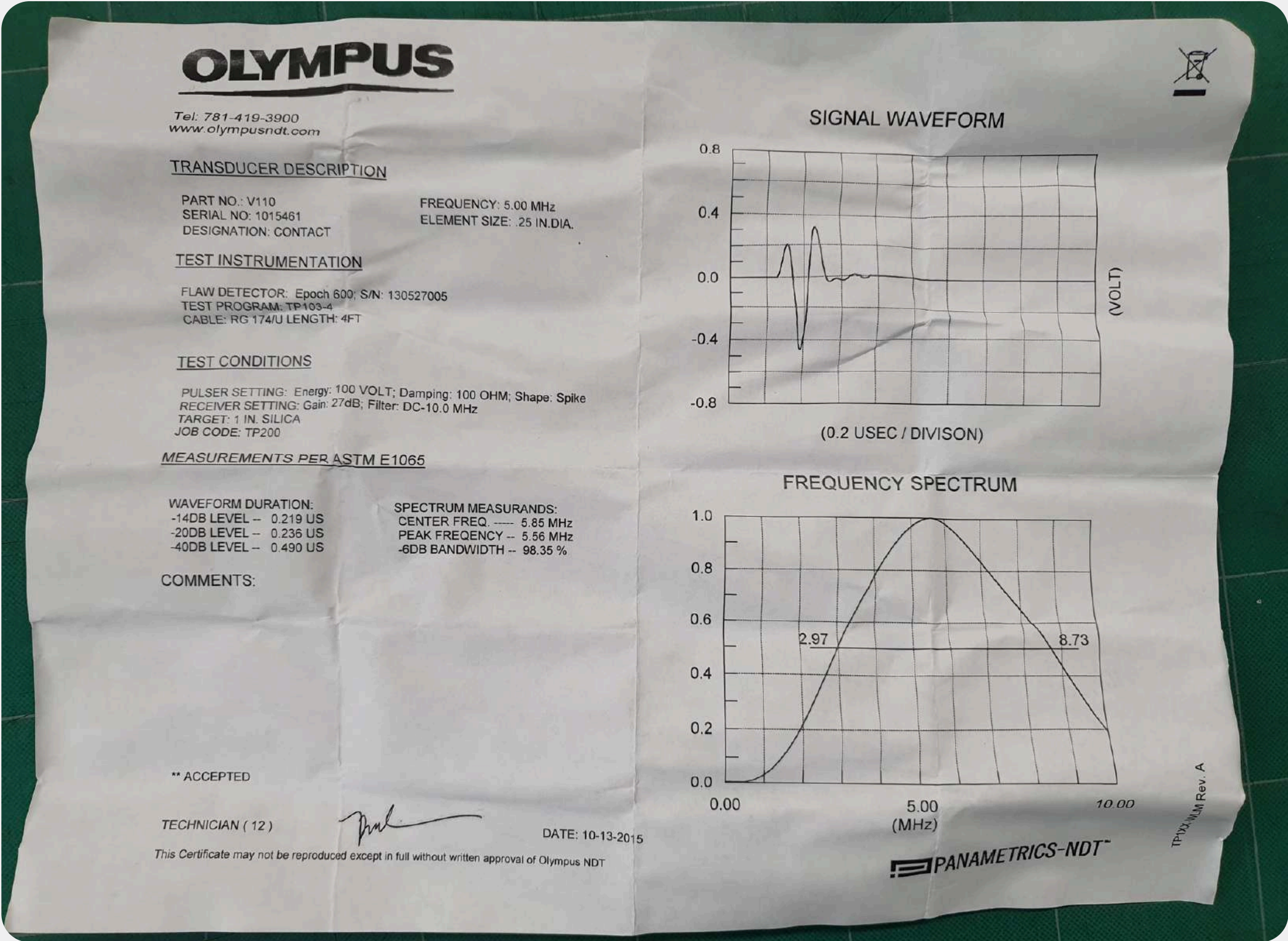
이 시편의 재료는 무엇인가?

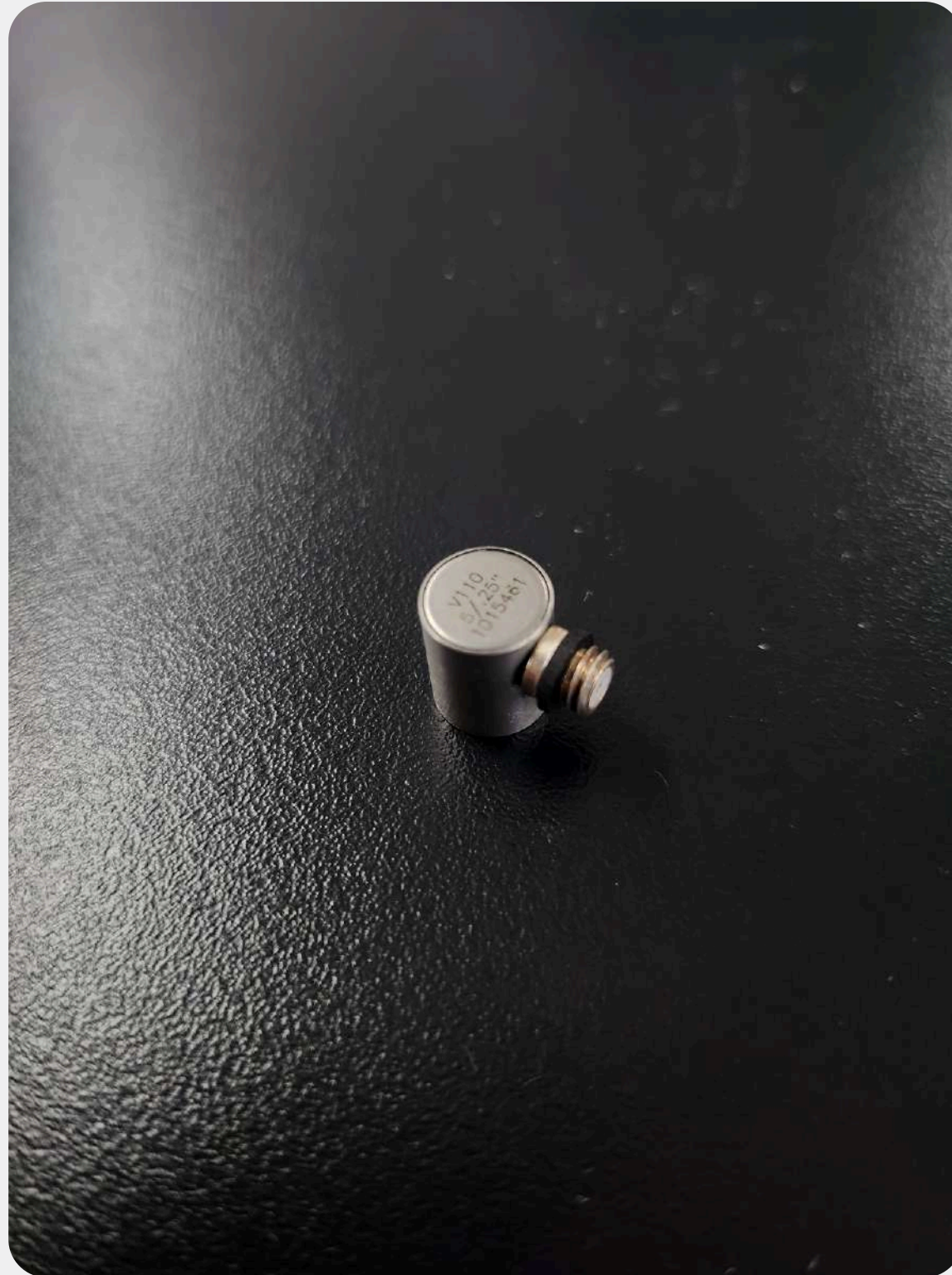
이 시편에서의 초음파 신호의 속력을 구할 수 있는가?











Thickness

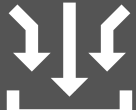
$$d = v \times \left(\frac{t}{2} \right)$$

$$\text{Thickness} = \frac{\text{Ultrasonic Velocity} \times \text{Round-Trip Time}}{2}$$

1. 탐촉자가 초음파 펄스를 송출 후 재료 통과
2. 반대편 표면에서 초음파 펄스 반사
3. 반사된 초음파 신호를 탐촉자가 수신
4. 송출 시간과 수신 시간의 차이를 측정
5. 두께 계산

 Velocity

$$v = \frac{d \times 2}{t}$$

 속도와 두께가 모두 입력값인 경우

버니어 캘리퍼스로 측정 후 오차범위 설정, 두께 측정

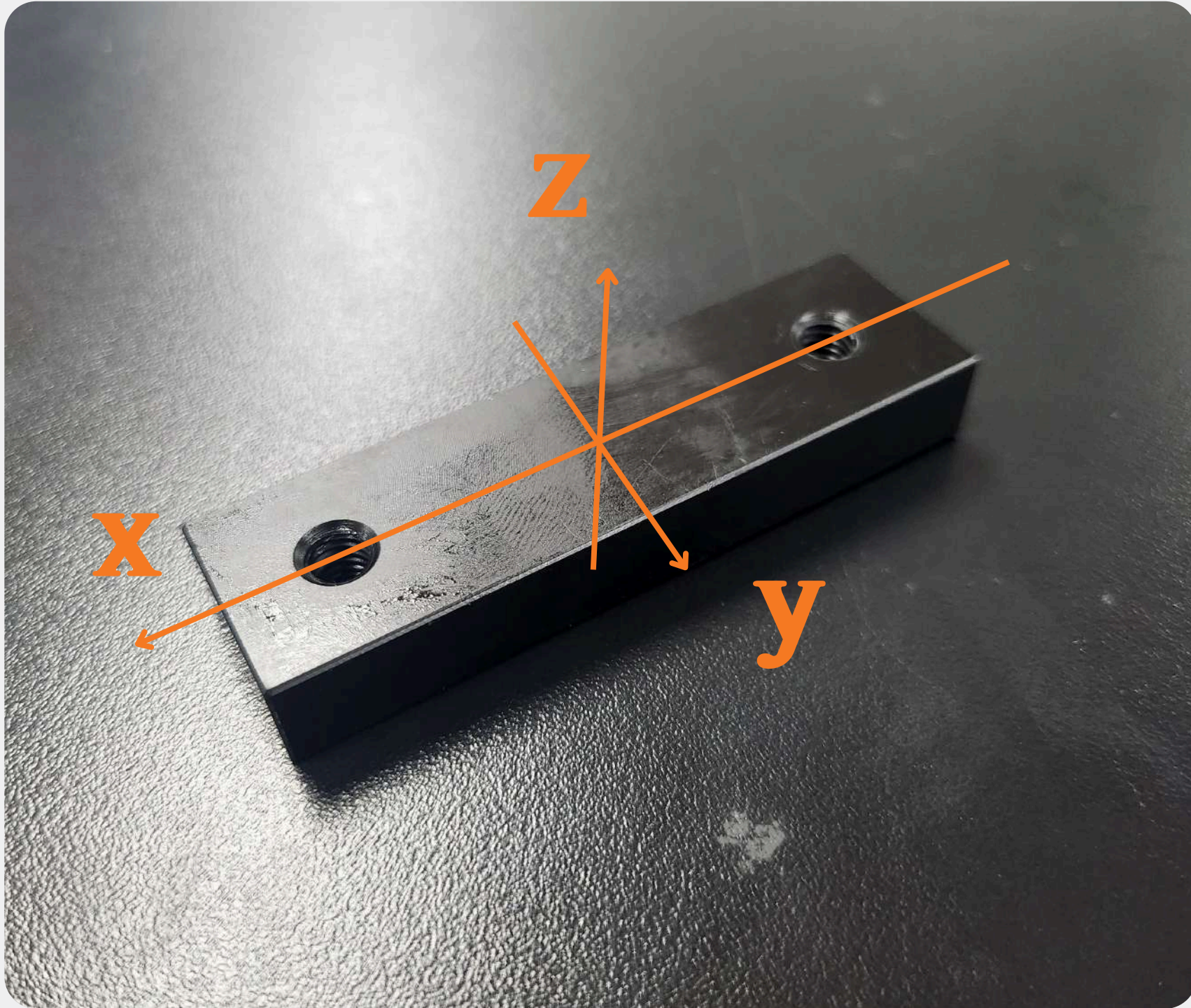
뒷면에서 반사되어 이동한 총 거리와 시간, 속도 계산

초음파의 매질속도로부터 물질의 재료를 예측가능

 Velocity

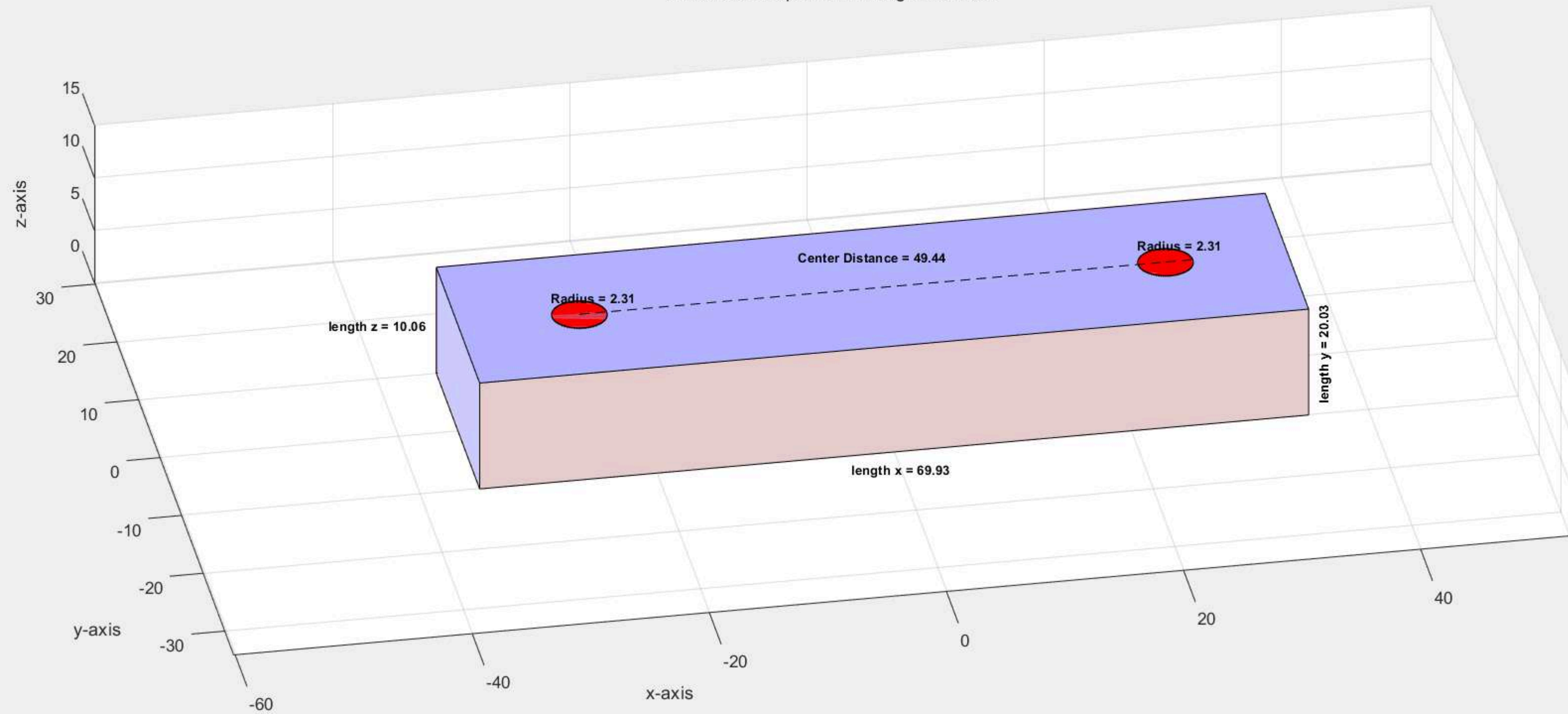
$$v = \frac{d \times 2}{t}$$

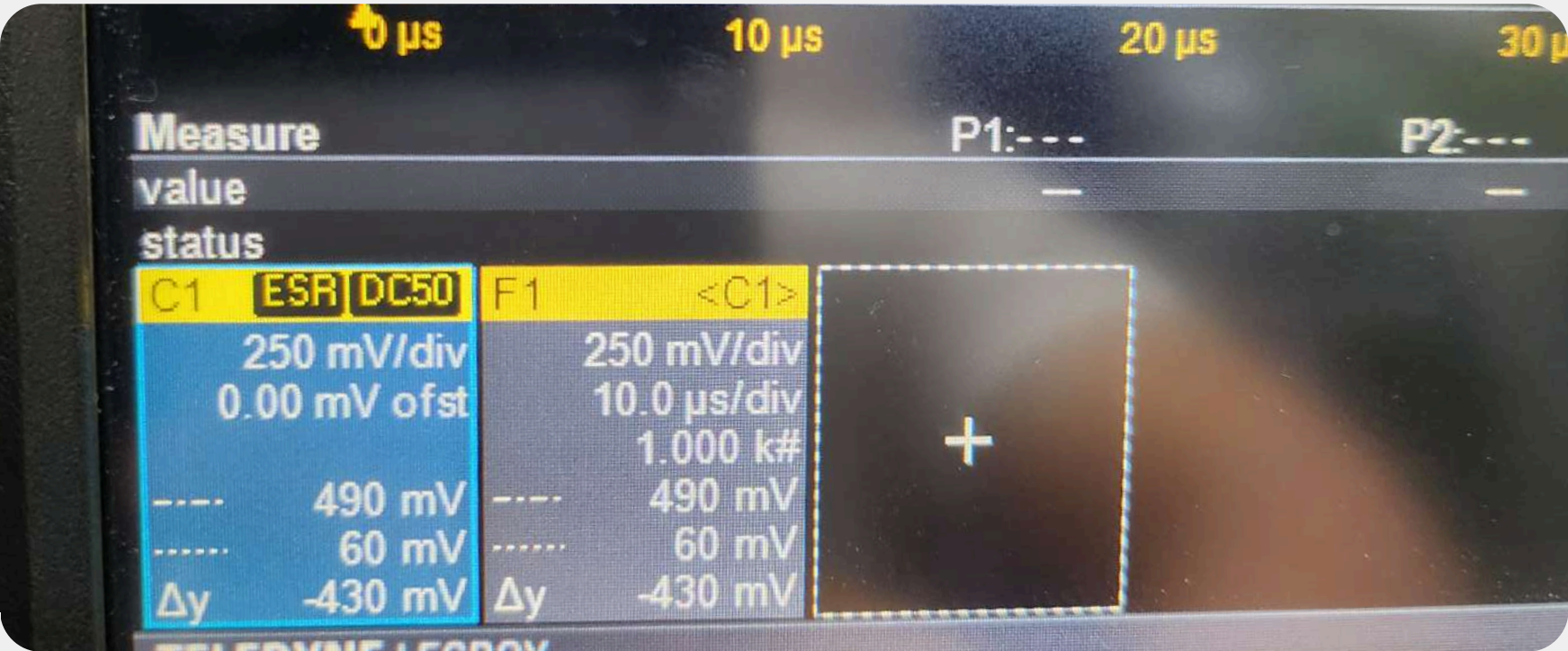
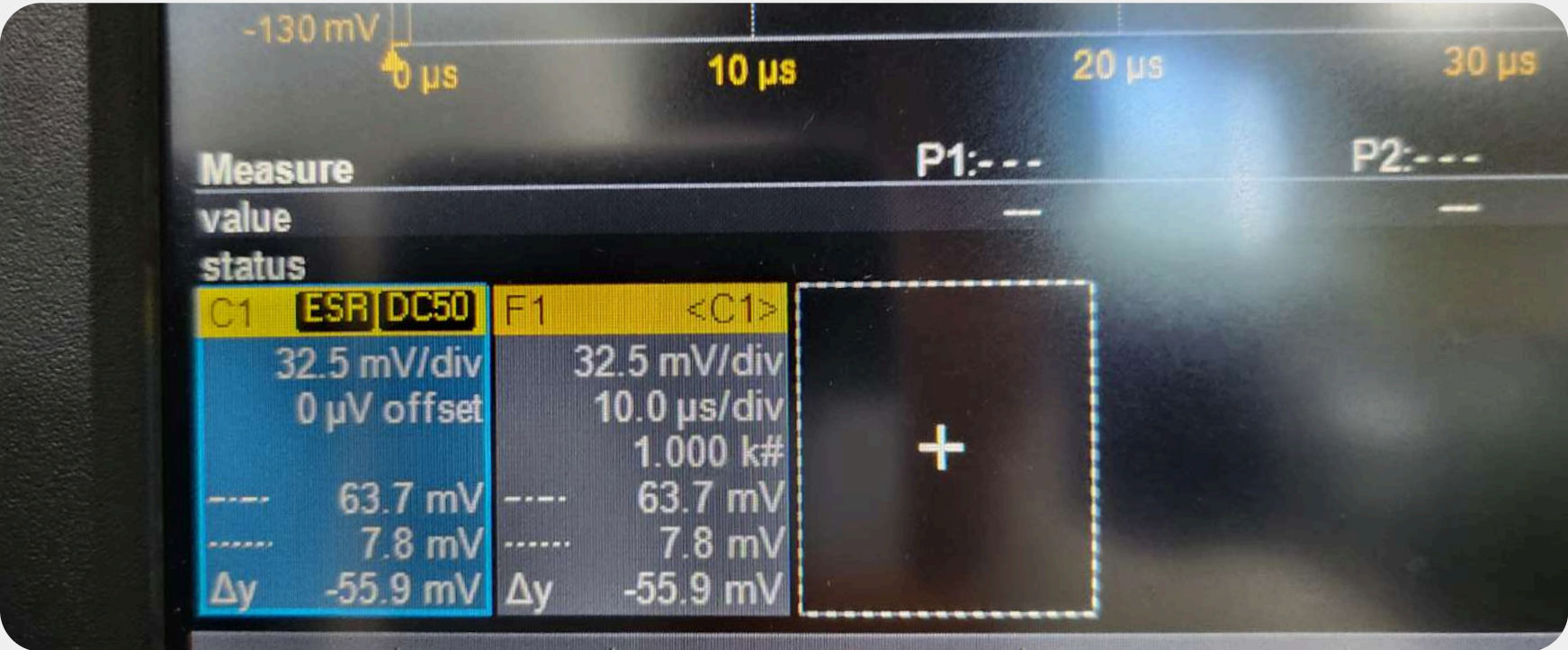
Material (Menu Option)	Longitudinal Velocity		Min Range*		Max Range*	
	in/us	m/s	in	mm	in	mm
SS 303 (SS303)	0.223	5664	0.040	1.02	5.000	127.00
STEEL 1018 (Steel)	0.233	5918	0.040	1.02	5.000	127.00
Al 2024 (Alum)	0.251	6375	0.050	1.27	5.000	127.00
PLEXIGLAS (Plexi)	0.106	2692	0.040	1.02	2.000	50.80
PVC (PVC)	0.094	2388	0.035	0.89	2.000	50.80
POLYSTYRENE (Poly)	0.081	2057	0.035	0.89	2.000	50.80



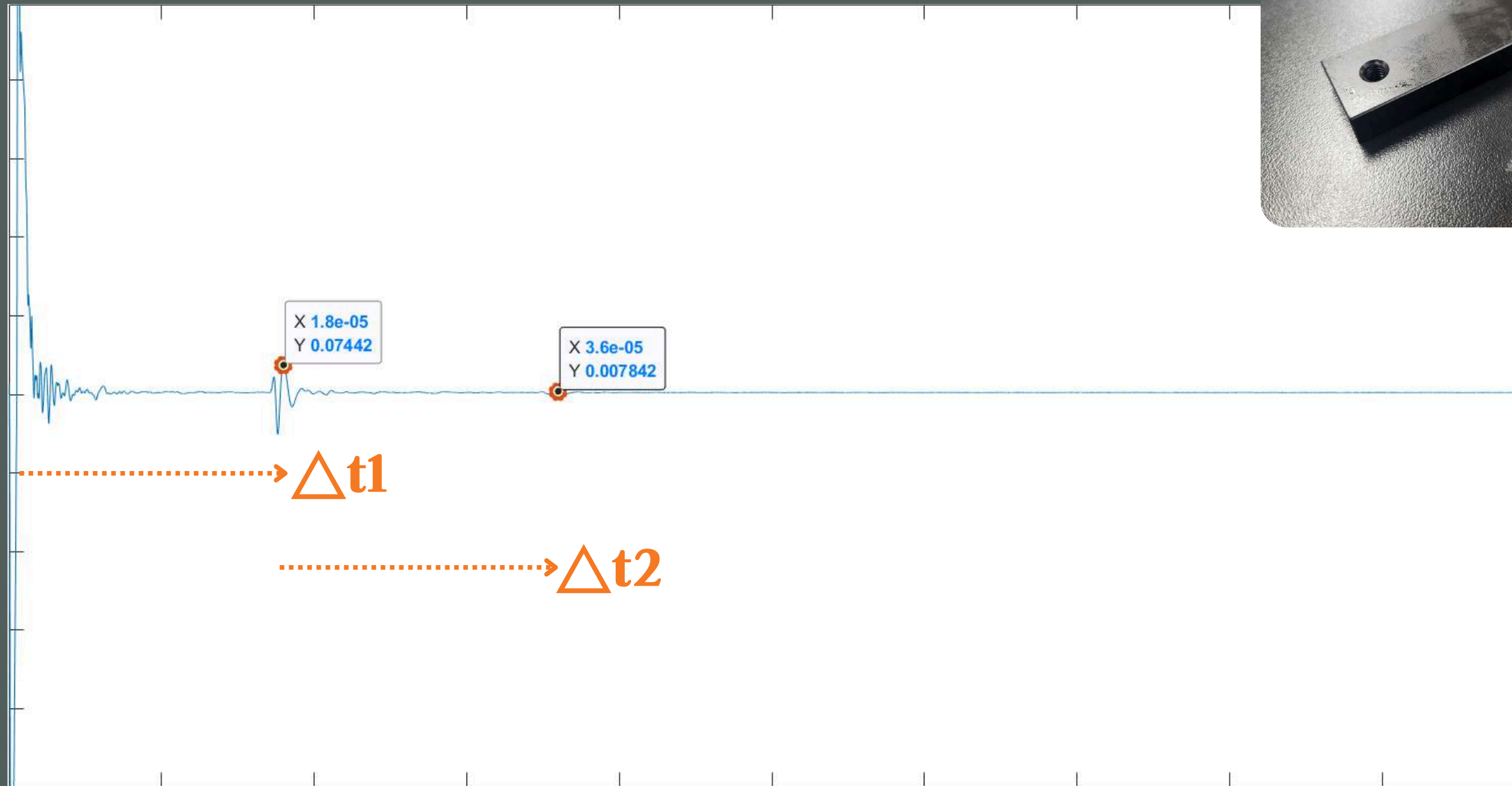
- x-axis : 69.93mm
- y-axis : 20.03mm
- z-axis : 10.06mm
- Hole r : 2.31mm
- Hole center distance : 49.44mm

3D cuboid Shape with Through two holes



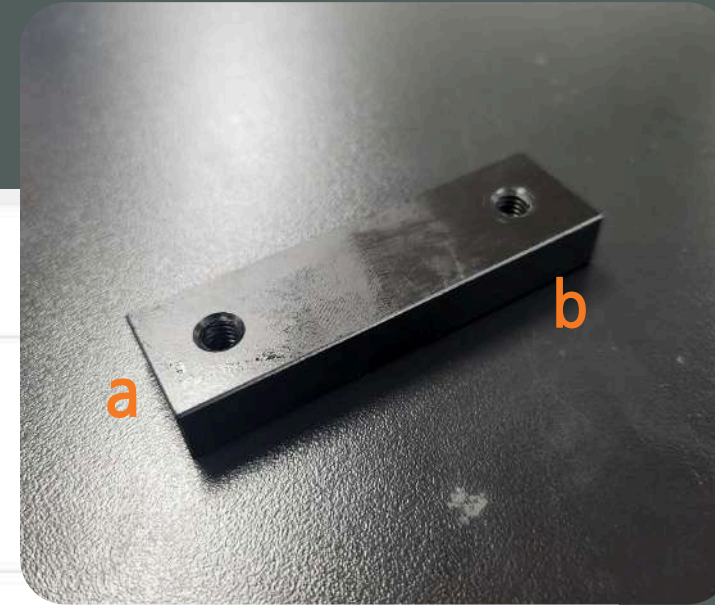
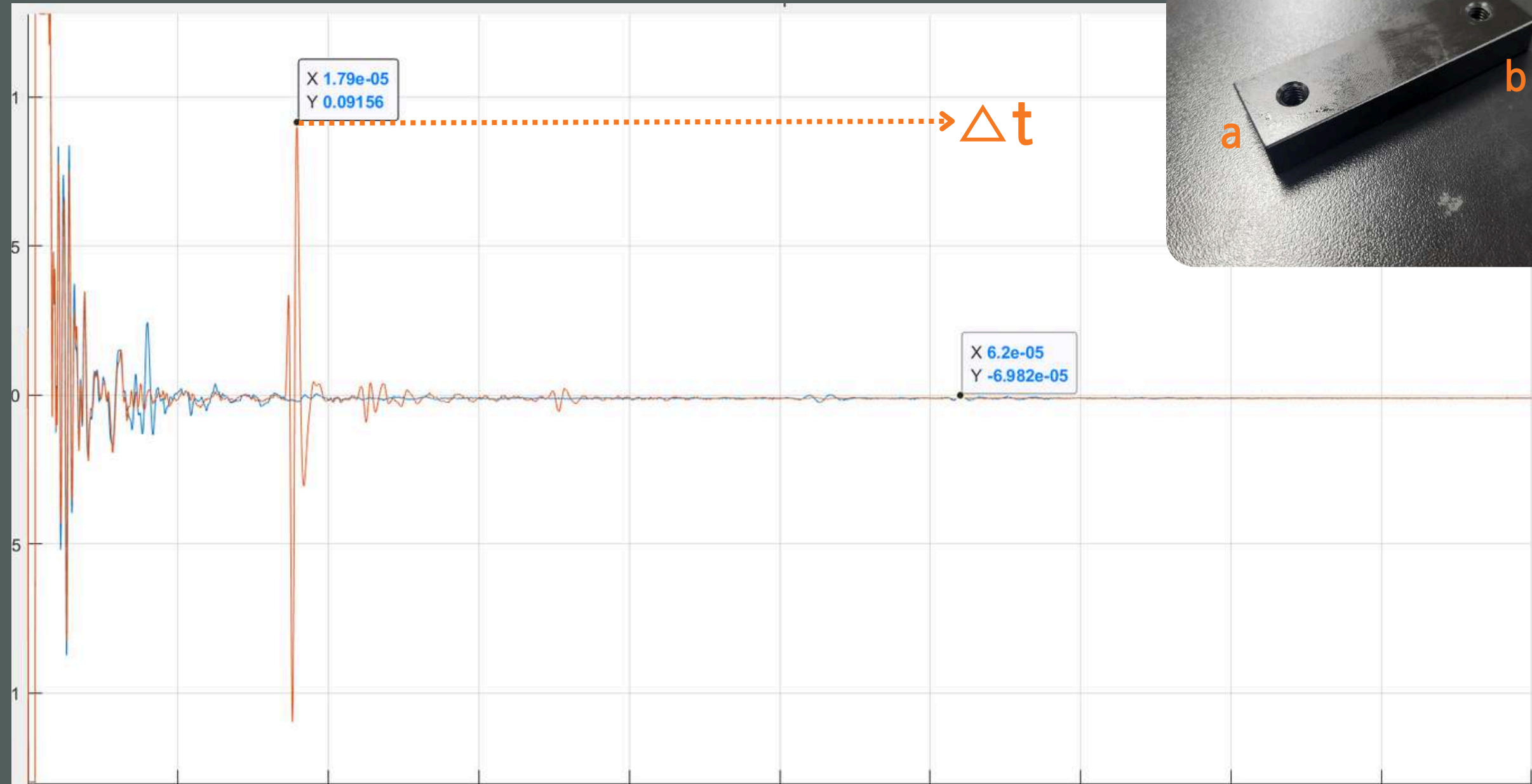


Method 1

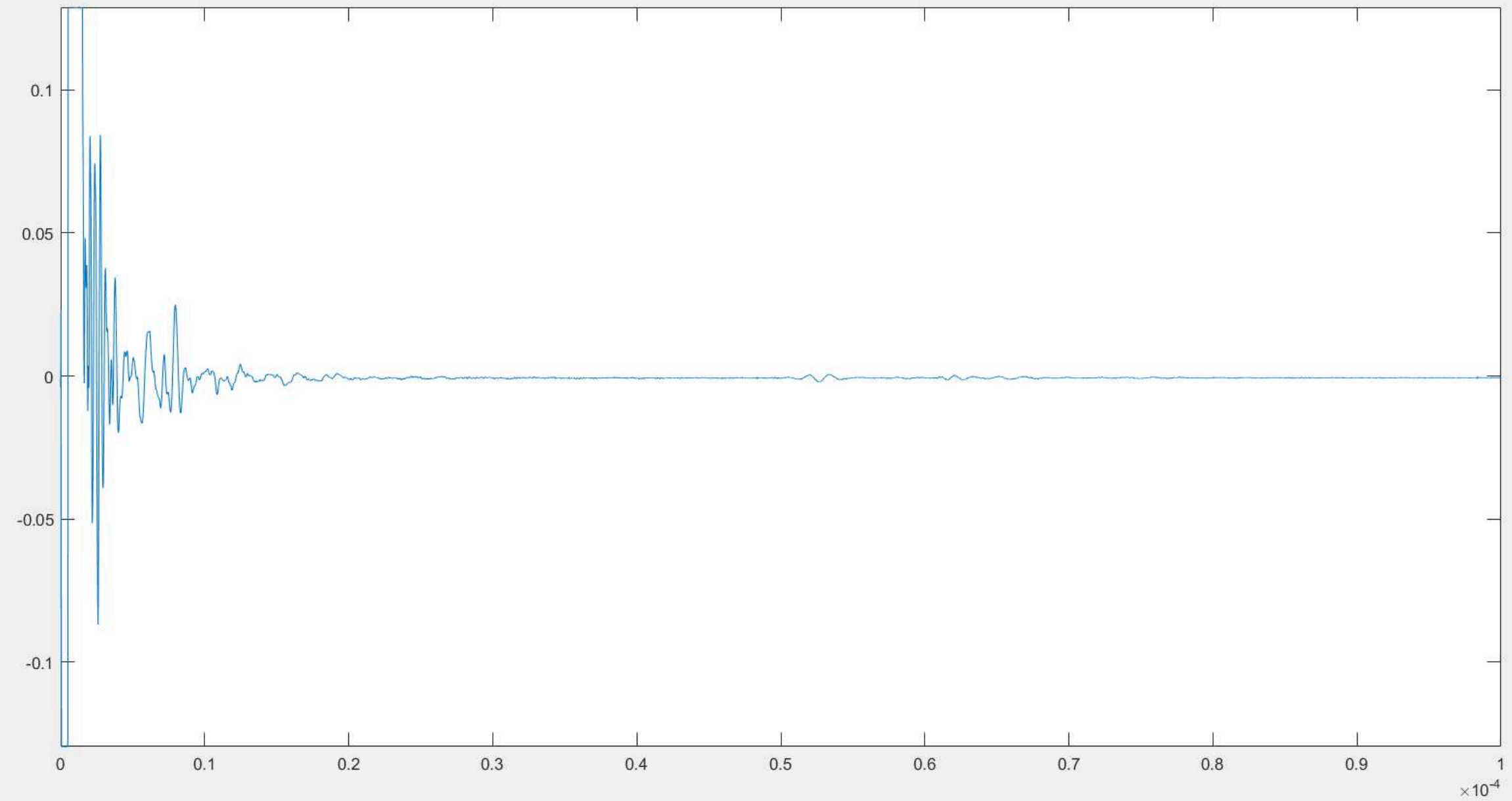


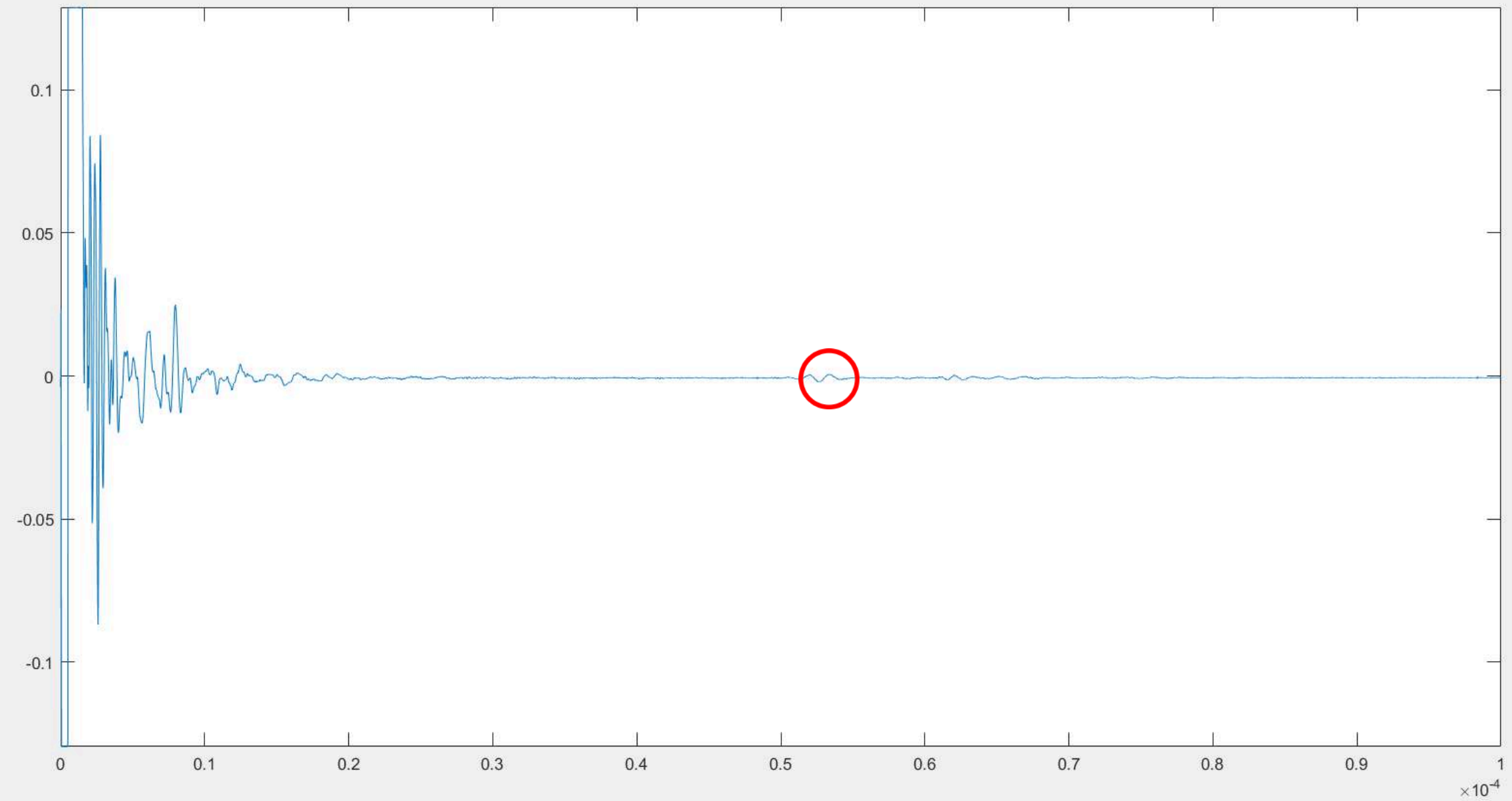
$$v_{\text{avg}} = \frac{\left(\frac{2d}{\Delta t_1}\right) + \left(\frac{2d}{\Delta t_2}\right)}{2}$$

Method 2



$$v = 2 \times \frac{(a-b)}{\Delta t}$$

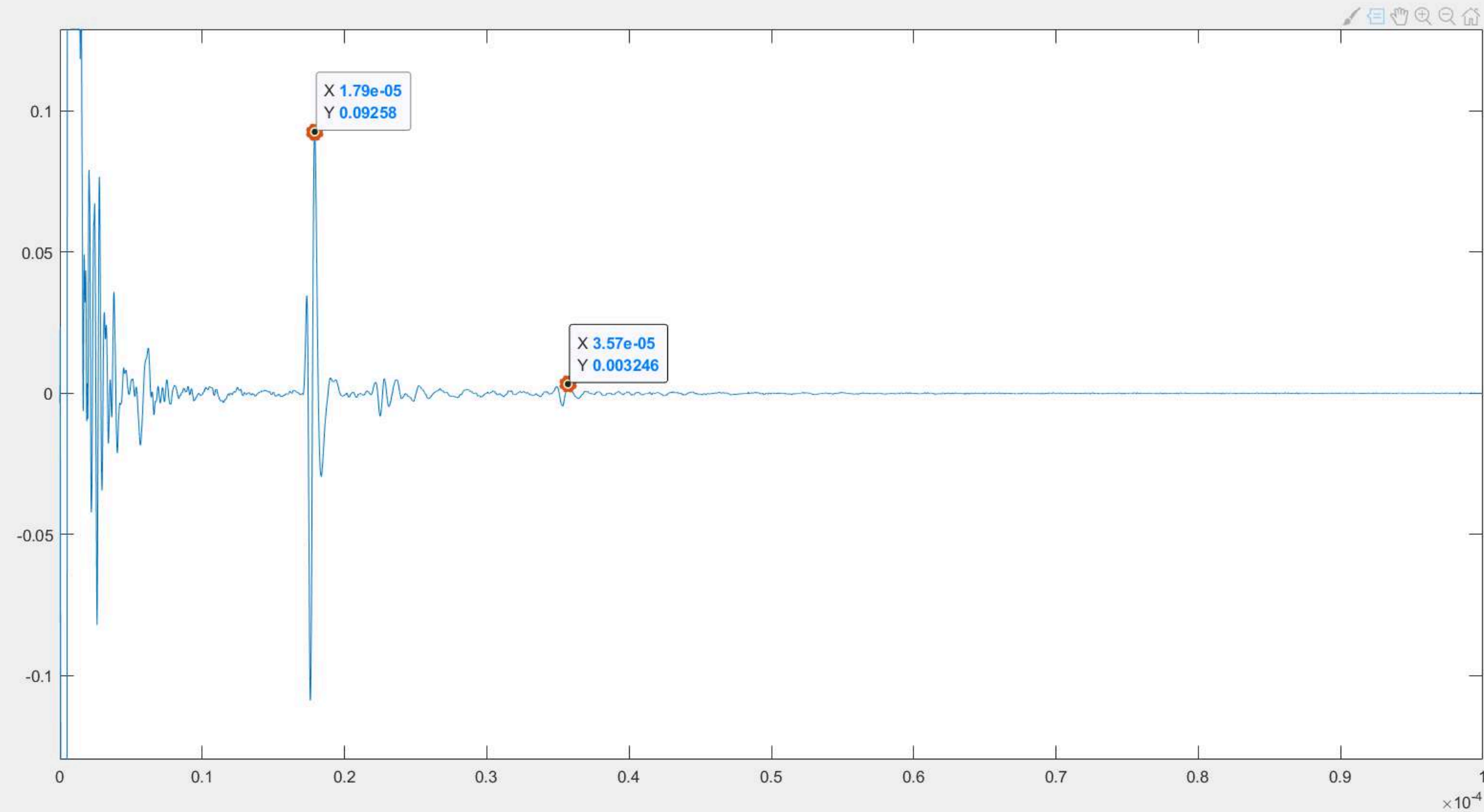


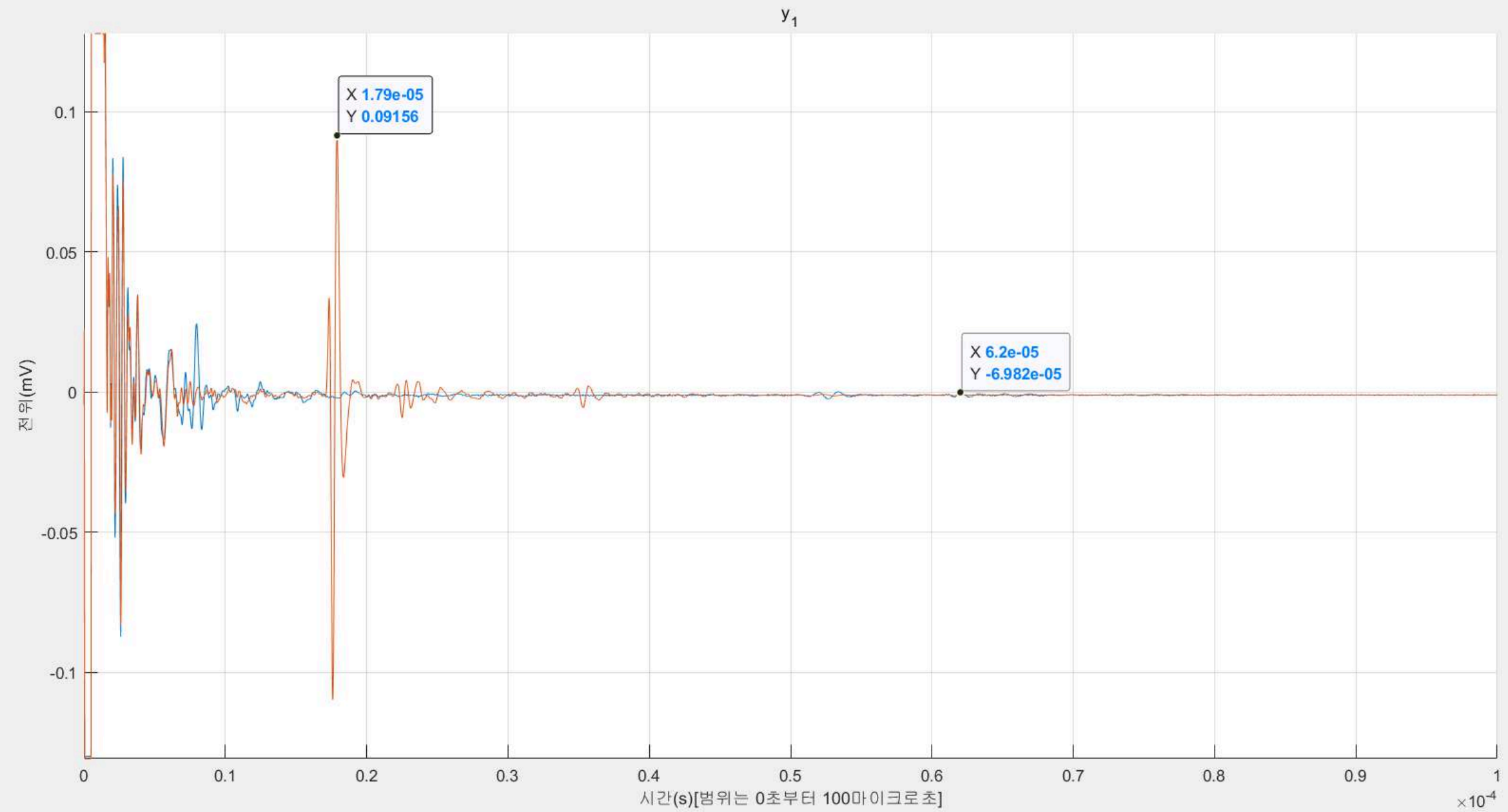




i : 0.0000179s

ii : 0.0000357s



 x1 & y1

 x1 & y1

$$v1_{(x1y1)} = \frac{2 \cdot (69.93 \text{ mm} - 20.03 \text{ mm}) \cdot 0.001 \text{ m/mm}}{0.00005335 \text{ s} - 0.0000179 \text{ s}}$$

1. 거리를 변환하고 차이를 구합니다:

$$69.93 \text{ mm} - 20.03 \text{ mm} = 49.90 \text{ mm}$$

$$49.90 \text{ mm} \times 0.001 \text{ m/mm} = 0.04990 \text{ m}$$

2. 시간의 차이를 계산합니다:

$$0.00005335 \text{ s} - 0.0000179 \text{ s} = 0.00003545 \text{ s}$$

$$v1_{(x1y1)} = \frac{2 \cdot 0.04990 \text{ m}}{0.00003545 \text{ s}}$$

$$v1_{(x1y1)} = \frac{0.09980 \text{ m}}{0.00003545 \text{ s}}$$

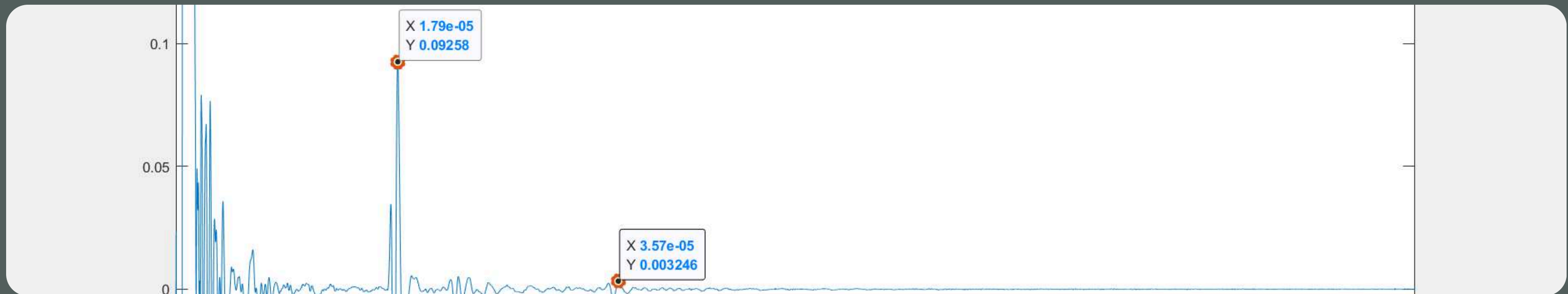
$$v1_{(x1y1)} \approx 2814.49 \text{ m/s}$$

 x1 & y1

y1의 첫번째 뒷면 반사파와 두번째 뒷면 반사파의 차이를 Δt 라 할 때,

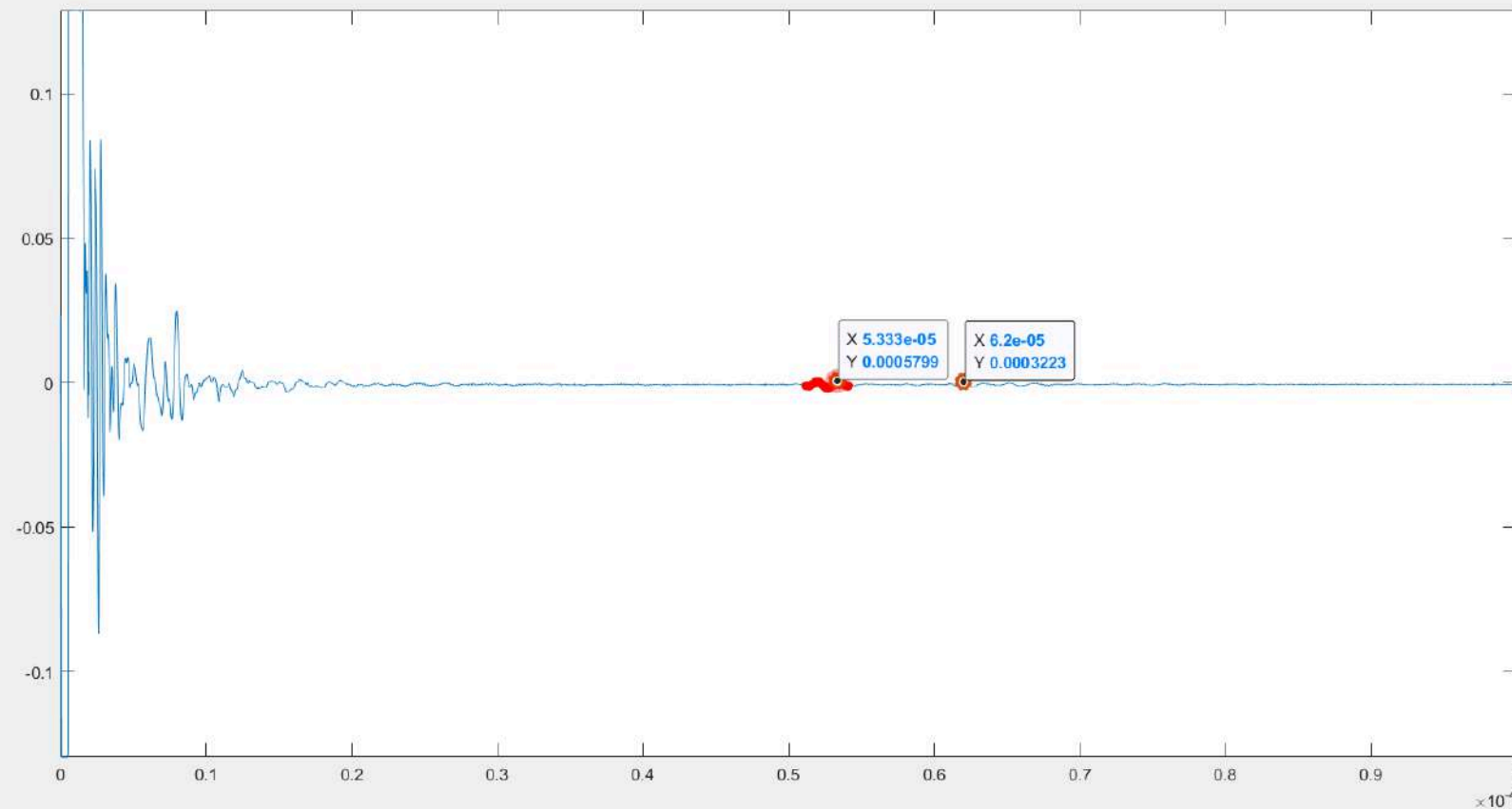
$$v_{y1} = \frac{2d}{\Delta t}$$

$$v_{y1} = \frac{2 \cdot (20.03 \text{ mm} \times 0.001 \text{ m/mm})}{0.0000357 \text{ s} - 0.0000179 \text{ s}} = \frac{0.04006 \text{ m}}{0.0000178 \text{ s}} = 2250 \text{ m/s}$$



 **x1 & y1**

오차의 크기가 너무 큰 이유는?

 x1 & y1


x1은 ii의 경우가 뒷면에서 반사된 신호일 가능성이 높다

$$v2_{(x1y1)} = \frac{2 \cdot (69.93 \text{ mm} - 20.03 \text{ mm}) \cdot 0.001 \text{ m/mm}}{0.000062 \text{ s} \quad 0.0000179 \text{ s}}$$

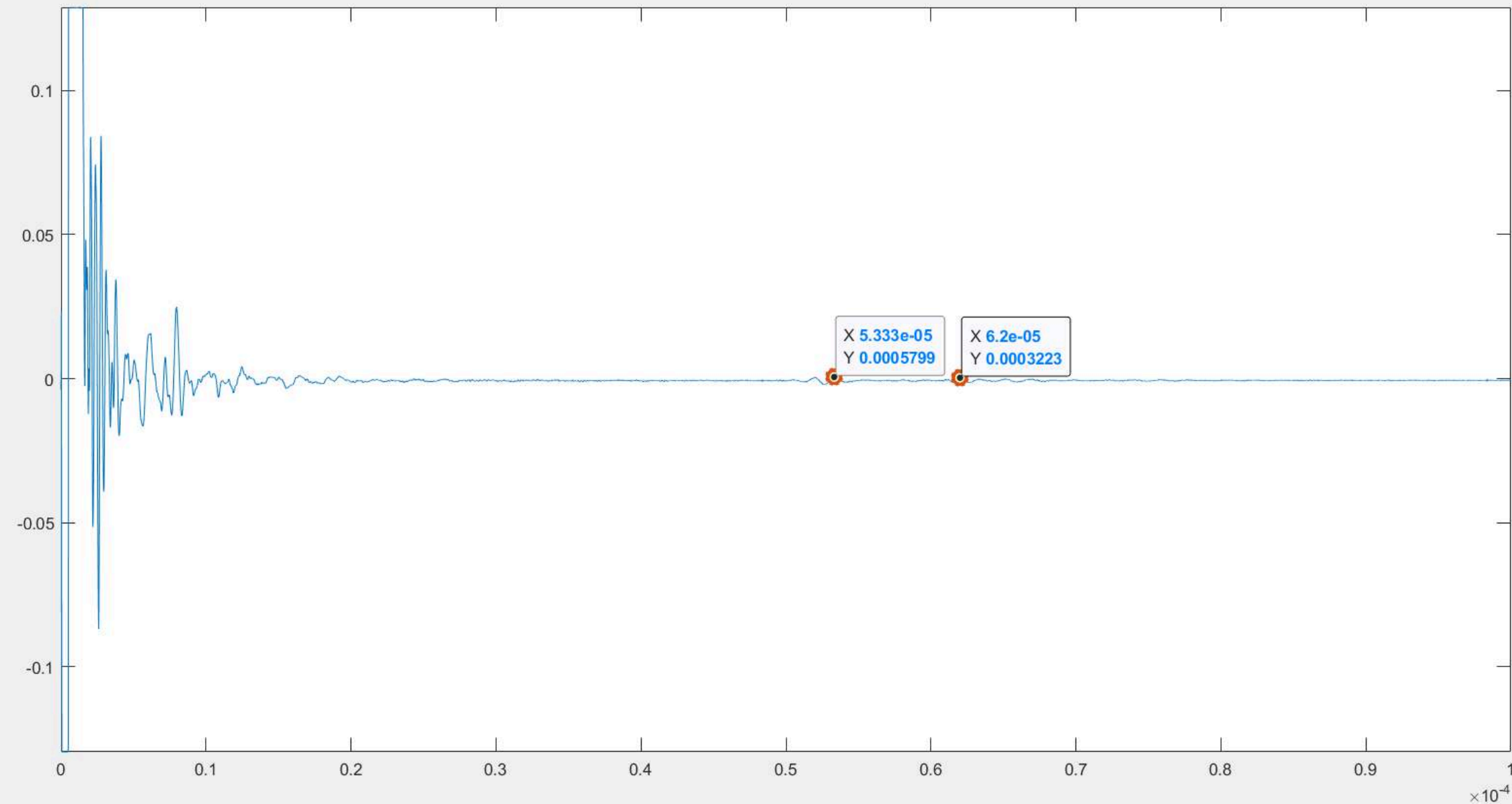
$$v2_{(x1y1)} = \frac{2 \cdot 0.04990 \text{ m}}{0.0000441 \text{ s}} = \frac{0.09980 \text{ m}}{0.0000441 \text{ s}}$$

$$v2_{(x1y1)} \approx 2263.05 \text{ m/s}$$

 x1 & y1

$$v_1 = \frac{2 \times 69.93 \times 0.001}{0.00005335}$$
$$v_1 = \frac{0.13986}{0.00005335}$$
$$v_1 \approx 2622.93 \text{ m/s}$$

$$v_2 = \frac{2 \times 69.93 \times 0.001}{0.000062}$$
$$v_2 = \frac{0.13986}{0.000062}$$
$$v_2 \approx 2255.81 \text{ m/s}$$

i : 0.00005335**ii : 0.000062**

x1 & y1

~~$$v_1 = \frac{2 \times 69.93 \times 0.001}{0.00005335}$$

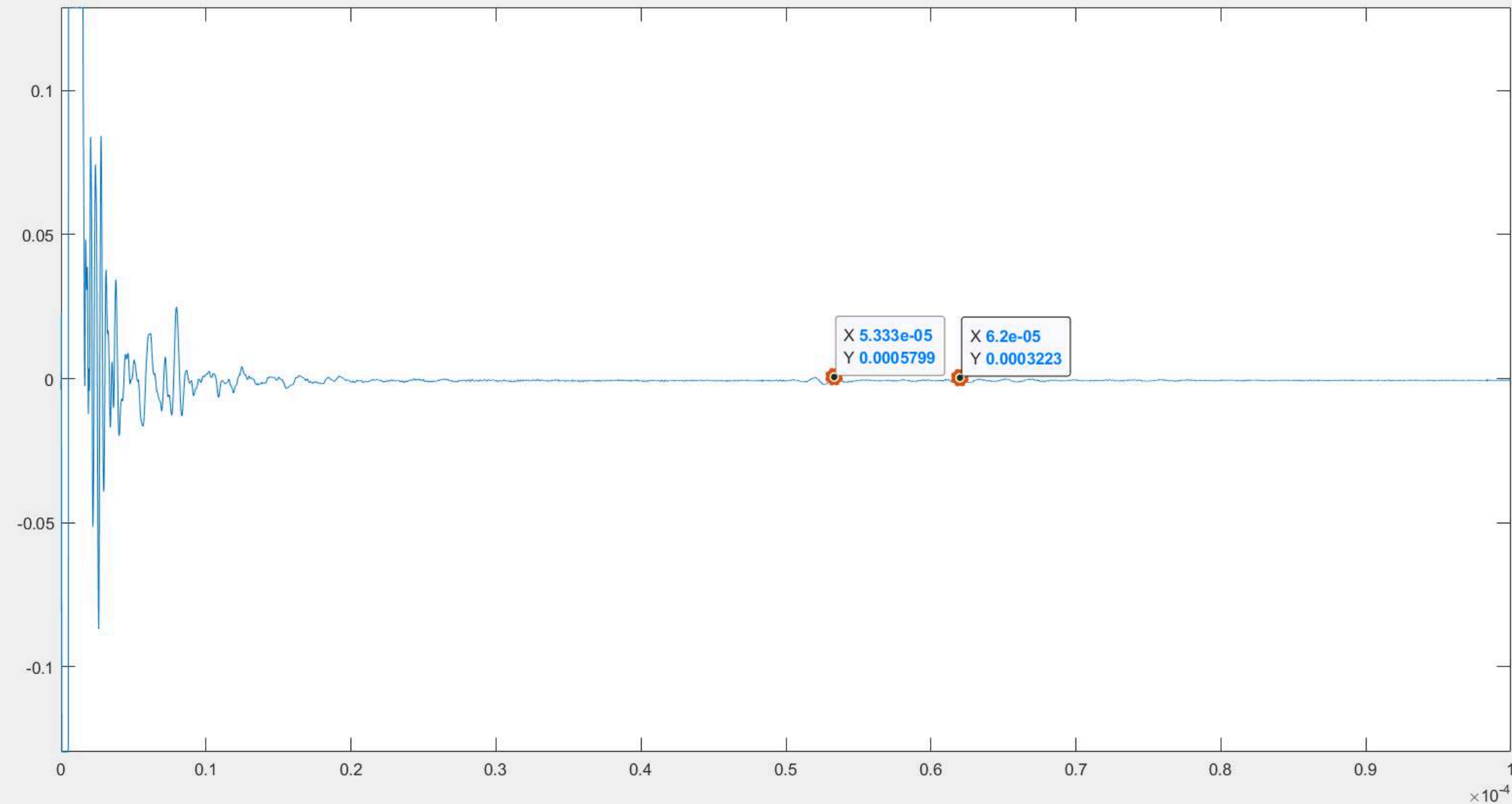
$$v_1 = \frac{0.13986}{0.00005335}$$

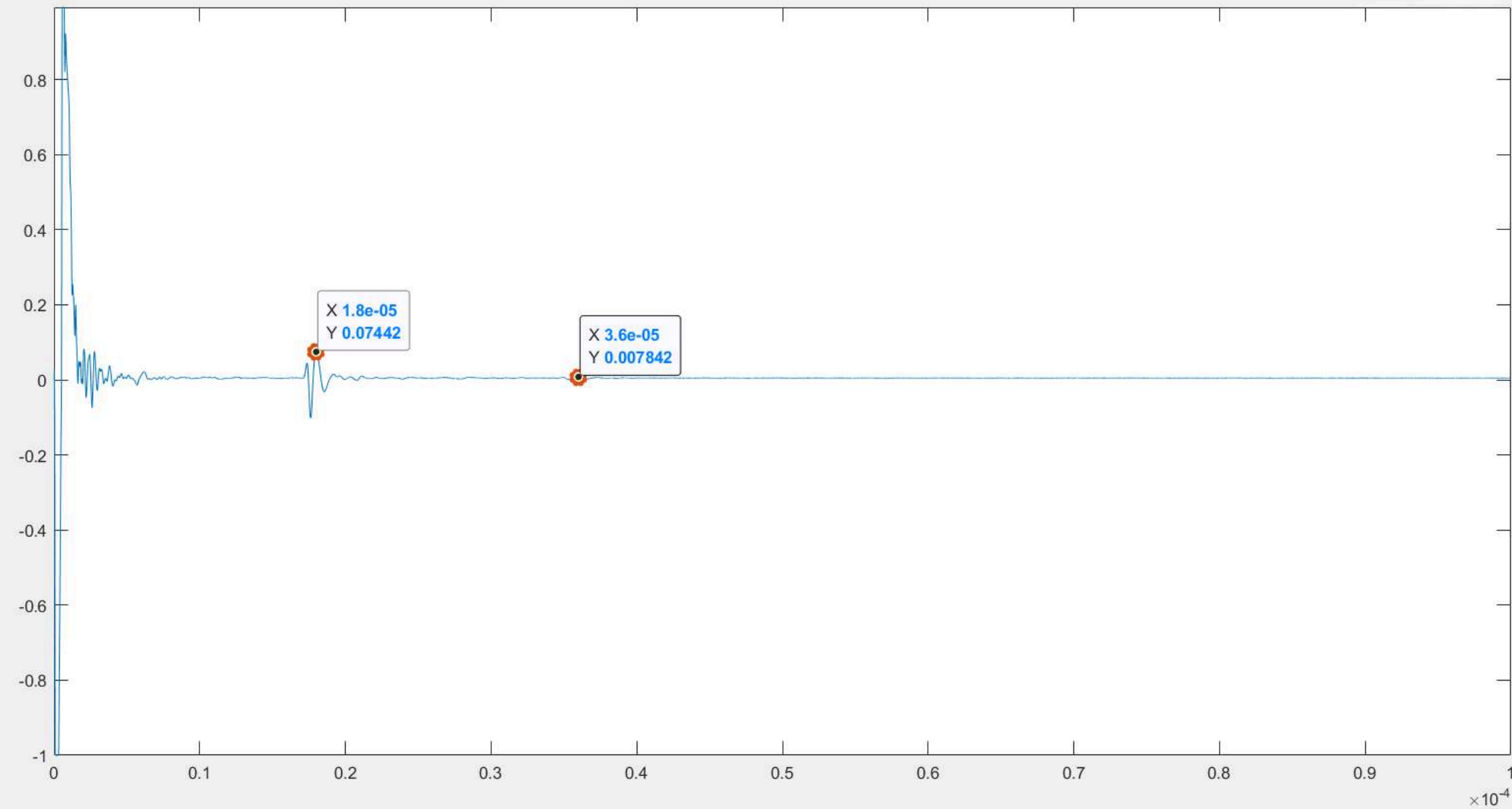
$$v_1 \approx 2622.93 \text{ m/s}$$~~

$$v_2 = \frac{2 \times 69.93 \times 0.001}{0.000062}$$

$$v_2 = \frac{0.13986}{0.000062}$$

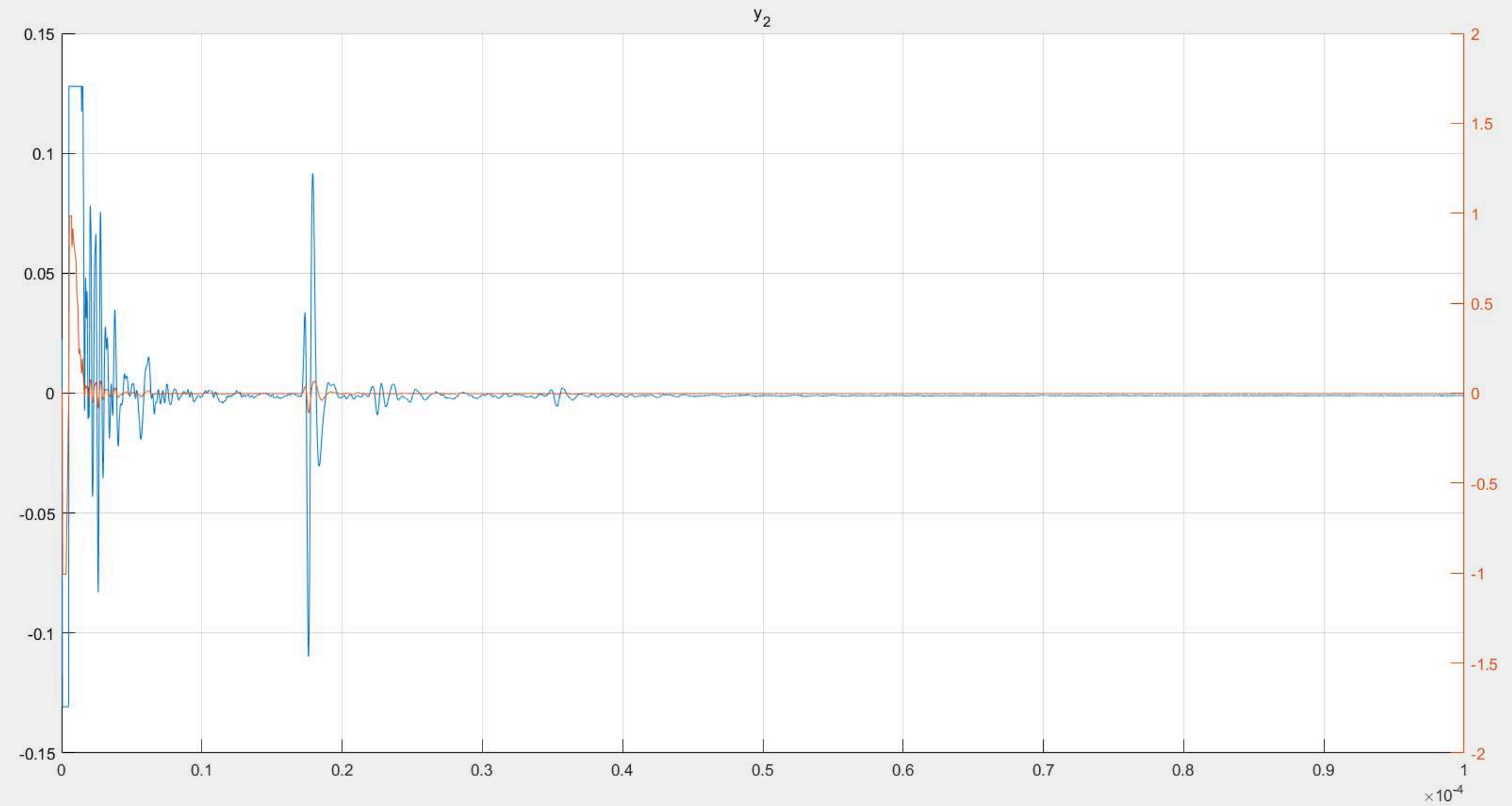
$$v_2 \approx 2255.81 \text{ m/s}$$

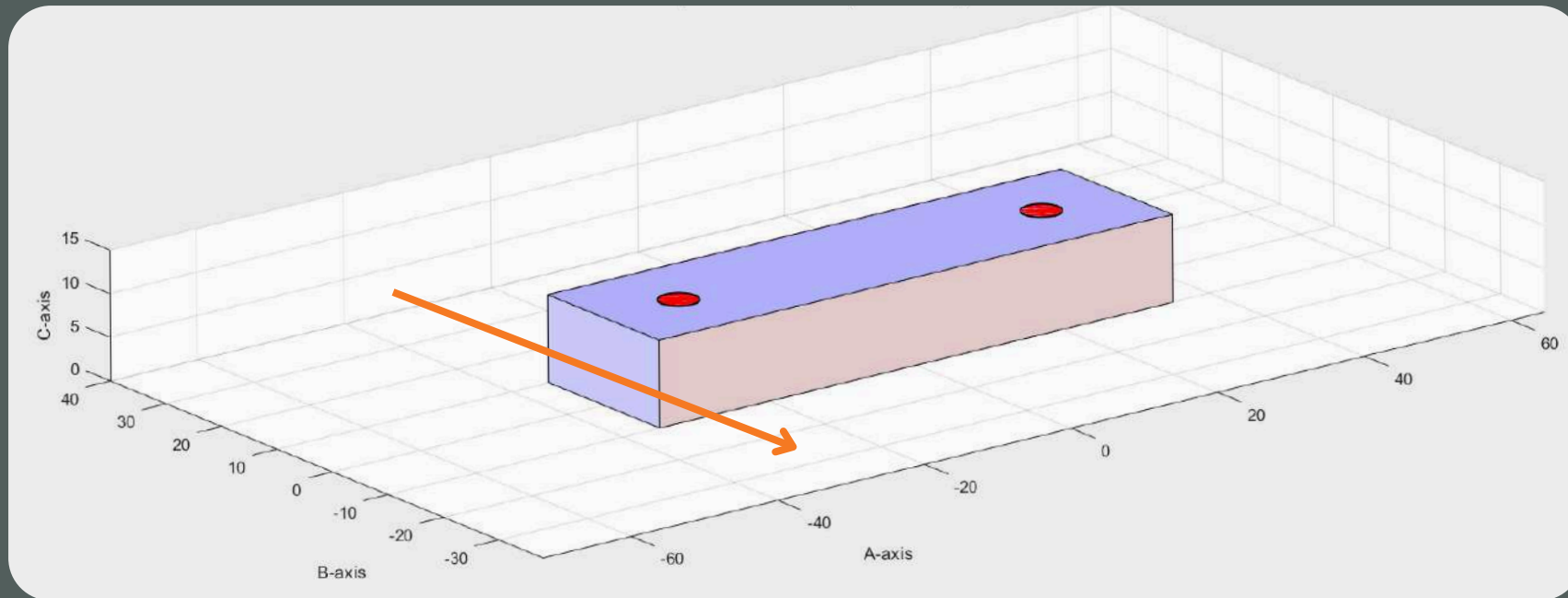
i : 0.00005335
ii : 0.000062


 y2**i : 0.000018s****ii : 0.000036s**

 y_1, y_2

x_1 과 z_2 사이의 매개변수





y2의 첫번째 반사신호와 두번째 반사신호 간의 시간 차 : 0.000018s
y길이는 20.03mm

$$v_{y2} = \frac{2d}{t}$$

$$v_{y2} = \frac{0.04006 \text{ m}}{0.000018 \text{ s}}$$

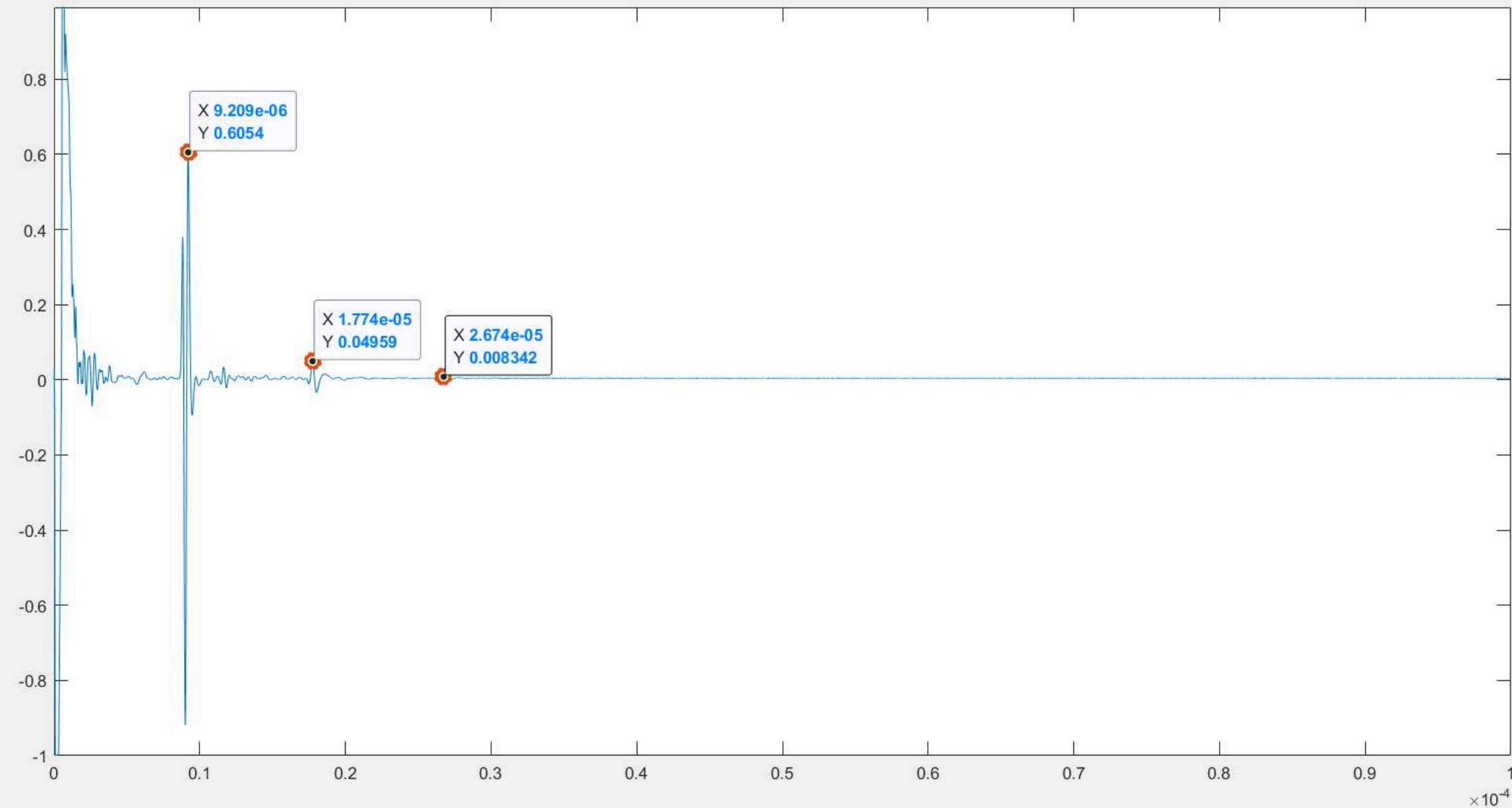
$$v_{y2} = 2225.56 \text{ m/s}$$

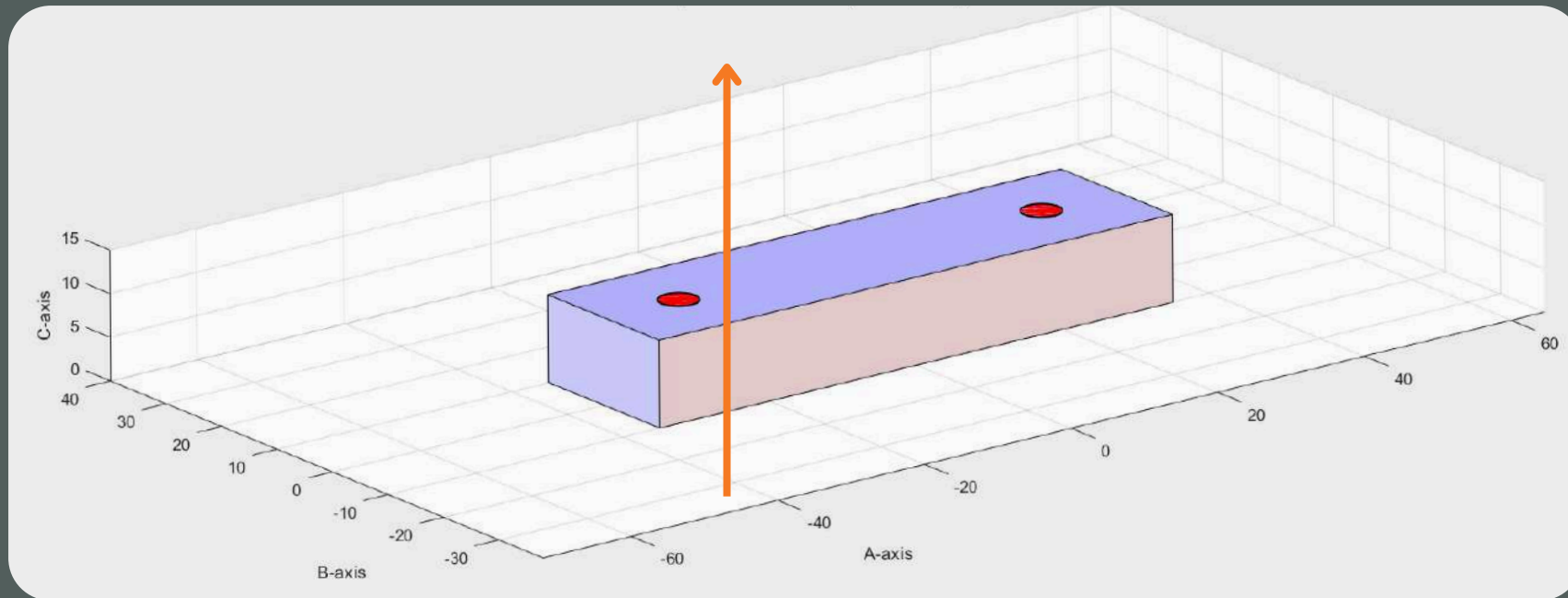


i : 0.000009209s

ii : 0.00001774s

iii : 0.00002674s





z2의 첫번째 반사신호와 두번째 반사신호 간의 시간 차 : 0.000008531s

z2의 첫번째 반사신호와 두번째 반사신호 간의 시간 차 : 0.000009s

z길이는 10.06mm

$$v1_{z2} = \frac{2 \times 0.01006 \text{ m}}{0.000008531 \text{ s}}$$

$$v1_{z2} \approx 2358.33 \text{ m/s}$$

$$v2_{z2} = \frac{2 \times 0.01006 \text{ m}}{0.000009 \text{ s}}$$

$$v2_{z2} \approx 2235.56 \text{ m/s}$$

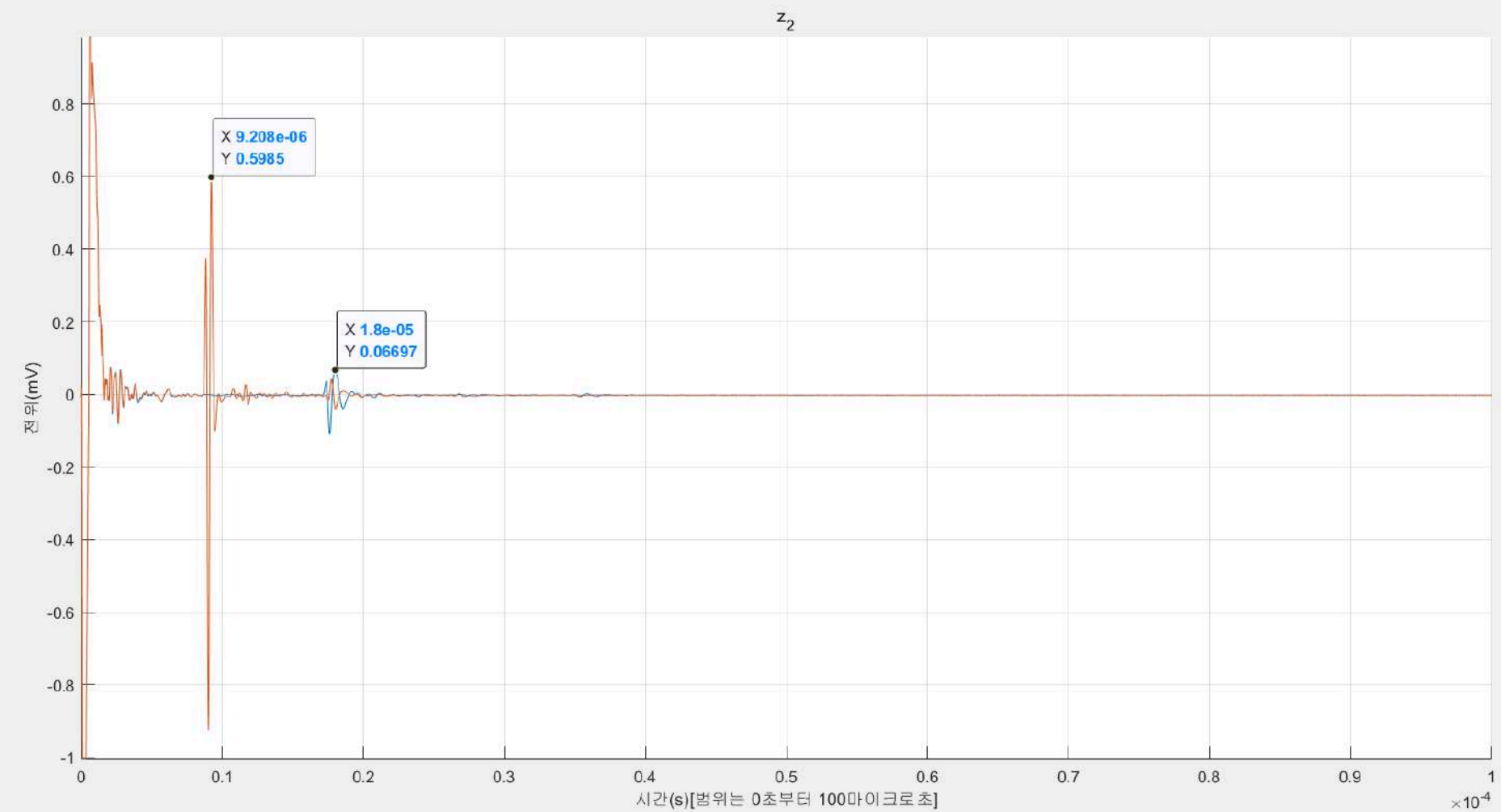
$$v_{\text{avg}} = 2296.945 \text{ m/s}$$

y2 & z2

$$v_{y2z2} = 2 \times \frac{(20.03 \text{ mm} - 10.06 \text{ mm}) \times 0.001 \text{ m/mm}}{0.000018 \text{ s} - 0.000009208 \text{ s}}$$

$$v_{y2z2} = 2 \times \frac{0.00997 \text{ m}}{0.000008792 \text{ s}}$$

$$v_{y2z2} = 2 \times 1134.15 \text{ m/s} \approx 2268.30 \text{ m/s}$$

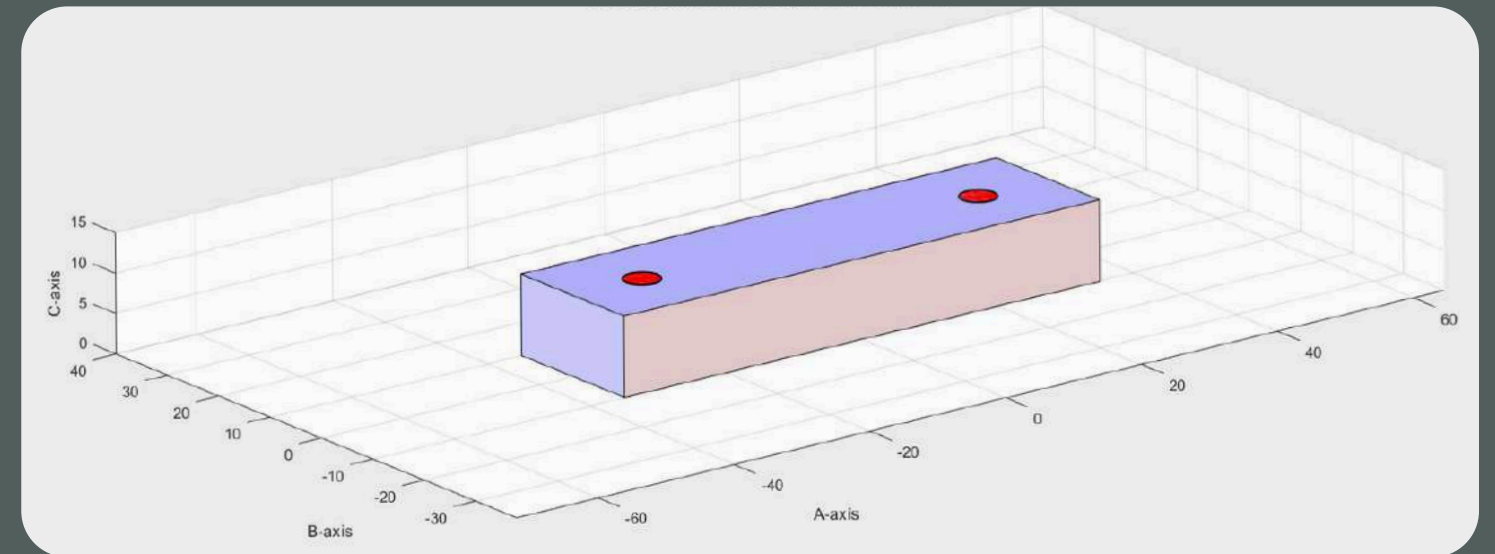


 z2 & x1

$$v_{z2x1} = 2 \times \frac{(69.93 \text{ mm} - 10.06 \text{ mm}) \times 0.001 \text{ m/mm}}{0.000062 \text{ s} - 0.000009208 \text{ s}}$$

$$v_{z2x1} = 2 \times \frac{0.05987 \text{ m}}{0.000052792 \text{ s}}$$

$$v_{z2x1} = 2 \times 1134.15 \text{ m/s} \approx 2268.30 \text{ m/s}$$



 Material 예측

- $x1 = 2251.81 \text{ m/s}$
- $y1 = 2250 \text{ m/s}$
- $y2 = 2250 \text{ m/s}$
- $z2 = 2296.945 \text{ m/s}$
- $x1 / y1 = 2263.95 \text{ m/s}$
- $y2 / z2 = 2268.30 \text{ m/s}$
- $z2 / x1 = 2268.30 \text{ m/s}$

$$v_{avg} = 2264.19 \text{ m/s}$$

ABS (Acrylonitrile Butadiene Styrene) : 2230 ~ 2250 m/s

Polycarbonate : 2260 ~ 2300 m/s

PVC (Polyvinyl Chloride) : 2218 m/s

Polyethylene oxide : 2250 m/s

PSO (Polysulfone) : 2240 m/s

Vinyl, Rigid : 2230 m/s

Dimethyl pentene Polymer : 2200 m/s

PLA (Polylactic Acid) : 2200–2300 m/s

 Material 예측

$$v_{avg} = 2264.19 \text{ m/s}$$

✓ ABS (Acrylonitrile Butadiene Styrene) : 2230 ~ 2250 m/s

Polycarbonate : 2260 ~ 2300 m/s

PVC (Polyvinyl Chloride) : 2218 m/s

Polyethylene oxide : 2250 m/s

PSO (Polysulfone) : 2240 m/s

Vinyl, Rigid : 2230 m/s

Dimethyl pentene Polymer : 2200 m/s

✓ PLA (Polylactic Acid) : 2200–2300 m/s

Material 예측

PLA (Polylactic Acid)



Material 예측

ABS (Acrylonitrile Butadiene Styrene)



Material 예측

PLA(Polylactic Acid)

옥수수 전분이나 사탕수수와 같은 재생 가능한 자원으로 만들어진 바이오 플라스틱

생분해성이 있어 환경에 상대적으로 친화적 시간이 지나면 자연적으로 분해

ABS(Acrylonitrile Butadiene Styrene)

석유 기반 화합물질로 만들어진 합성 플라스틱

생분해되지 않으며, 제조 및 처리 과정에서 유독 가스가 발생할 수 있음

PLA

Characteristics

- Extruder temperature: 190 - 220°C
- Bed temperature: 45 - 60°C
- Heated bed optional, no need for closed chamber
- Not heat resistant
- Easier to print
- Stronger and stiffer than ABS
- Made of renewable resources

Price

\$18 - 50+

*per 1kg spool

Producers/Manufacturers

- Polymaker
- MatterHackers
- Ultimaker
- Filamentum
- ColorFabb
- Zortrax

ABS

Characteristics

- Extruder temperature: 220 - 250°C
- Bed temperature: 95 - 110°C
- Heated bed required, closed chamber recommended
- Heat and impact resistant
- More durable and flexible, better mechanical properties
- Oil-based polymer

Price

\$18 - 50+

*per 1kg spool

Producers/Manufacturers

- Polymaker
- MatterHackers
- Ultimaker
- Hatchbox
- Flashforge
- Filamentum

